

DS Lab 10: Data Visualization III

Iris Dataset Analysis

Aim:

Load the Iris dataset, analyze its features, and perform visualizations using histograms and box plots to identify distributions and outliers.

```
In [2]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris

sns.set_style('whitegrid')
```

```
In [3]: # Load Iris dataset
iris = load_iris()
df = pd.DataFrame(iris.data, columns=iris.feature_names)

# Add species column
df['species'] = iris.target

df.head()
```

```
Out[3]:
```

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	species
0	5.1	3.5	1.4	0.2	0
1	4.9	3.0	1.4	0.2	0
2	4.7	3.2	1.3	0.2	0
3	4.6	3.1	1.5	0.2	0
4	5.0	3.6	1.4	0.2	0

```
In [4]: # Data information
df.info()
df.describe()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
#   Column                Non-Null Count  Dtype
---  -
0   sepal length (cm)      150 non-null    float64
1   sepal width (cm)       150 non-null    float64
2   petal length (cm)      150 non-null    float64
3   petal width (cm)       150 non-null    float64
4   species                150 non-null    int64
dtypes: float64(4), int64(1)
memory usage: 6.0 KB
```

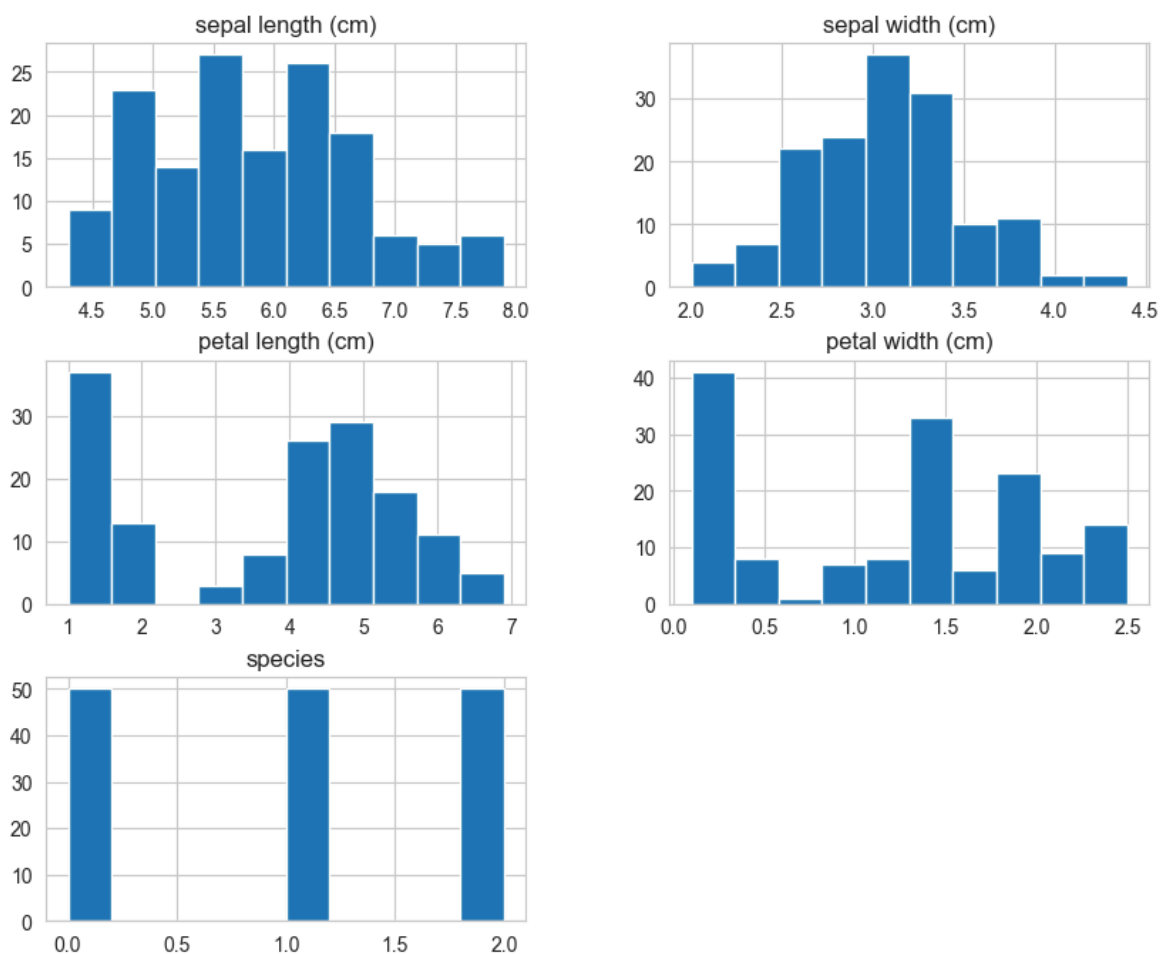
Out[4]:

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	species
count	150.000000	150.000000	150.000000	150.000000	150.000000
mean	5.843333	3.057333	3.758000	1.199333	1.000000
std	0.828066	0.435866	1.765298	0.762238	0.819232
min	4.300000	2.000000	1.000000	0.100000	0.000000
25%	5.100000	2.800000	1.600000	0.300000	0.000000
50%	5.800000	3.000000	4.350000	1.300000	1.000000
75%	6.400000	3.300000	5.100000	1.800000	2.000000
max	7.900000	4.400000	6.900000	2.500000	2.000000

Histograms of Features

```
In [5]: df.hist(figsize=(10,8))
plt.suptitle("Feature Distributions (Histograms)")
plt.show()
```

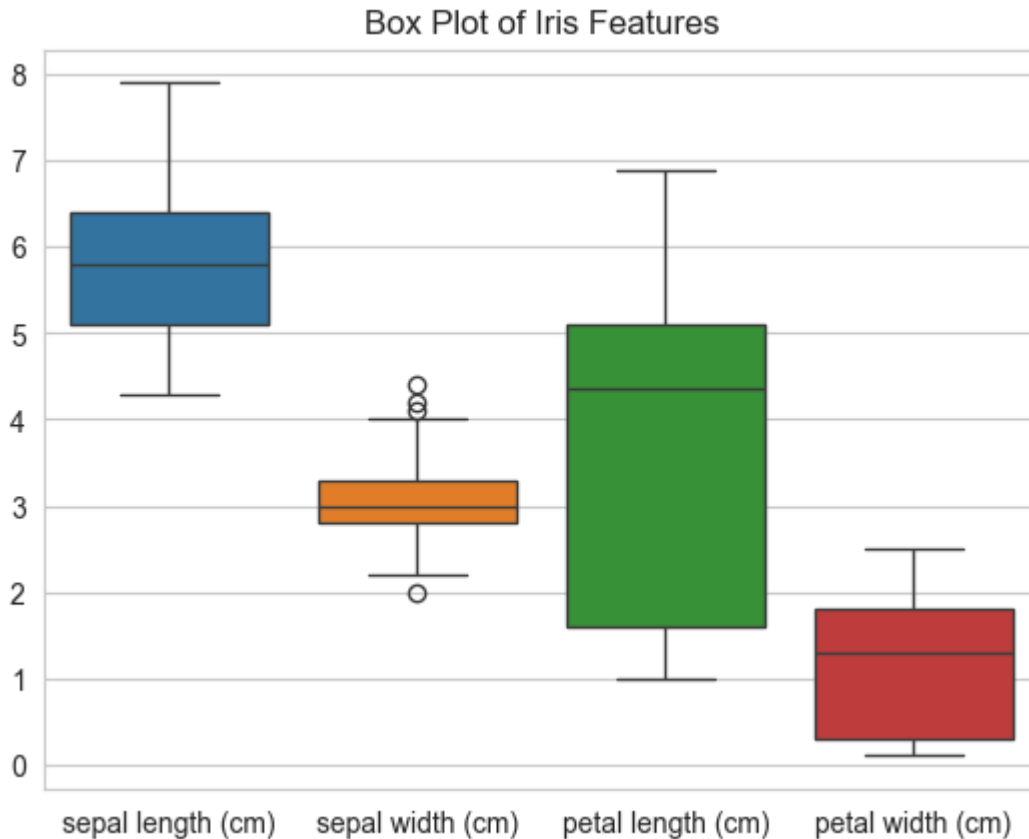
Feature Distributions (Histograms)



Insight: Petal features show clearer separation between species compared to sepal features.

Box Plot (All Features)

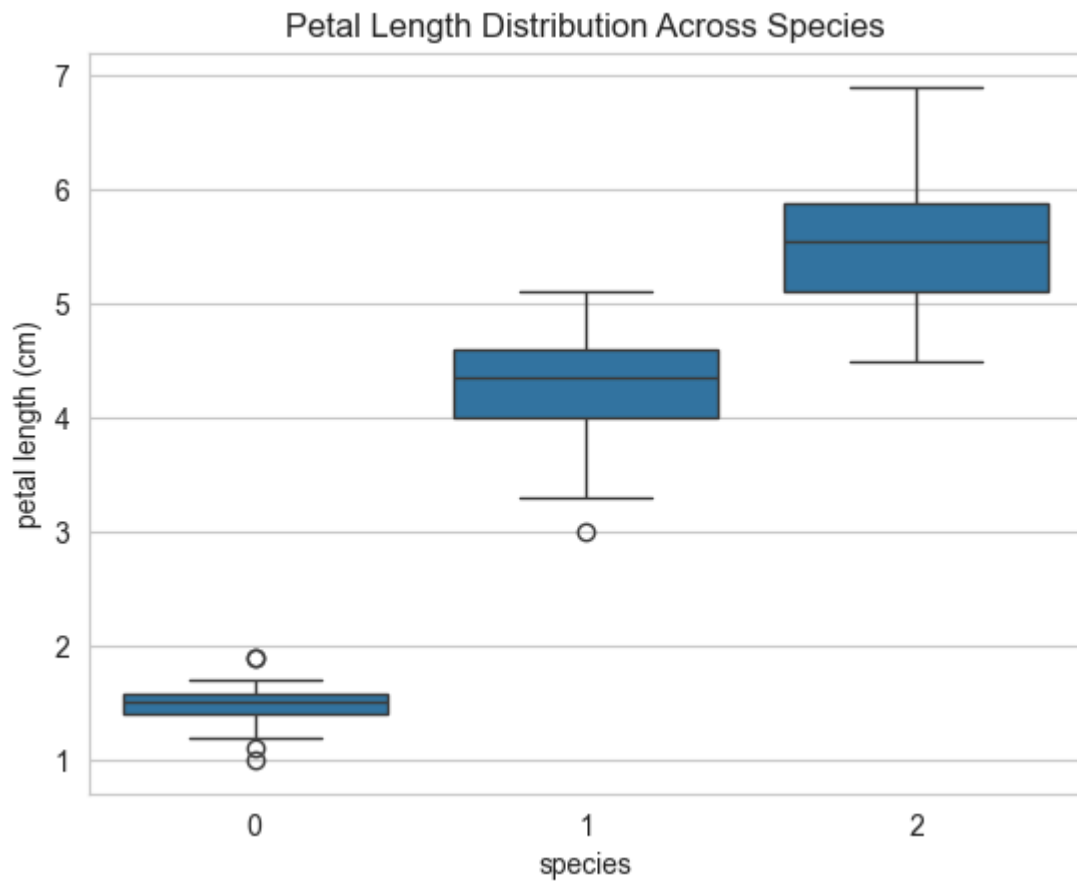
```
In [6]: sns.boxplot(data=df.drop('species', axis=1))  
plt.title("Box Plot of Iris Features")  
plt.show()
```



Insight: Some outliers are visible, especially in sepal width, while petal features are more consistent.

Improved Box Plot (Species vs Petal Length)

```
In [7]: sns.boxplot(x='species', y='petal length (cm)', data=df)  
plt.title("Petal Length Distribution Across Species")  
plt.show()
```



```
In [8]: df['species'] = df['species'].map({0: 'Setosa', 1: 'Versicolor', 2: 'Virginica'})
```

Insight: Petal length clearly separates species, making it a strong feature for classification.

Conclusion

The Iris dataset visualization shows that petal-related features provide better separation between species and help in identifying patterns and outliers.